

ActiveWAM: Supplementary Material

Anonymous Authors

APPENDIX

This appendix provides the implementation settings needed to interpret the method and the additional benchmark results. ActiveWAM uses the same model, sensor inputs, and action interface throughout each matched comparison. All numerical fields in the reported configuration and evaluation are specified below. Existing numerical results retain their original provenance.

A. Training Configuration

Training uses eight A100 80GB GPUs. The frozen video prior is used only to prepare transformed histories; deployment receives raw RGB, robot state, language, and measured camera metadata. The temporal history length satisfies the Wan VAE grouping constraint. Invalid boundary frames are masked rather than filled with future observations.

B. History Inversion Details

a) Training-time video prior and inverse direction.: The inversion operator is the frozen T2V path of Wan2.2-TI2V-5B [1]; the current image is not clamped as an I2V anchor. With data at $\sigma = 0$ and noise at $\sigma = 1$, an Euler inverse step is

$$\begin{aligned} U_{k+1} &= Z_k + (\sigma_{k+1} - \sigma_k)v_k, \\ v_k &\equiv v_\psi(Z_k, \sigma_k; \ell). \end{aligned} \quad (1)$$

Eight increasing integration steps end at $\tau_{\text{inv}} = 0.1$, measured in the data-to-noise coordinate of Eq. 1. Scheduler indices and the model’s timestep embedding are separate coordinates; neither is substituted directly for σ . We use classifier-free guidance scale 5. Direct decoding $R(U) = D_{\text{Wan}}(U)$ supplies the RGB bridge. A clean estimate $D_{\text{Wan}}(U - \sigma v_\psi(U, \sigma; \ell))$ and a denoising round trip are different readouts and require separate controls. Guided correction changes the path, so the procedure is not claimed to be an exact inverse.

b) Task weights and feature readout.: Let $P_{iu,j}^{bh}$ be the frozen source attention from video query (i, u) to non-padding task-text key j , in block b and head h . For selected blocks S_A and detached text-token weights α_j , define

$$a_{iu} = \frac{\sum_{b \in S_A, h, j} \alpha_j P_{iu,j}^{bh}}{|S_A|N_h}. \quad (2)$$

$$w_g(i, u) = \frac{a_{iu}}{\sum_{i', u'} m_{i' u'} a_{i' u'} + \epsilon}. \quad (3)$$

The query/key direction is visual-to-text. Each attention row is softmax normalized over all non-padding text keys before the selected subset is summed. The reported TTI models were trained using this full-key normalization. Matched legacy/full-key training controls give changes of +1.0 percentage point on RoboTwin-AV compound and +0.5 on the TAVIS history

diagnostic. These implementation checks are separate from the final benchmark scores. We use the selected task-text tokens, blocks $\{6, 12, 18, 24\}$, mean head aggregation, and source-probe $\sigma = 0.05$ throughout the paired transformation. The probe sees only the observed camera window and instruction. It does not supply segmentation ground truth or an unobserved task answer.

For readout scales $b \in S_F$, features are resampled to a common space-time grid, normalized per location, concatenated, and normalized again to give F_ψ . The final normalization is necessary for the unit-feature bounds used in the main text. The local scales are intended to retain small appearance cues; their inclusion alone does not demonstrate cue accuracy. Both source and transformed RGB use the same frozen probe and feature normalization. The attention blocks are $\{6, 12, 18, 24\}$, the feature scales use the same four blocks, and the source probe uses $\sigma = 0.05$. The selected task-text tokens have unit weight. The local readout uses the same frozen feature normalization for source and transformed RGB.

c) Correspondences and rejection.: Mutual nearest neighbors in source-feature cosine similarity propose temporal matches within each camera. A match is retained only when both endpoints are valid, its similarity passes γ_{match} , and its forward/backward cycle error is below γ_{cycle} . The confidence q_r is its cosine similarity mapped to $[0, 1]$. Source-only reliability filtering reduces mismatches; it is not proof of physical visibility. No matches are borrowed across cameras or future decision windows.

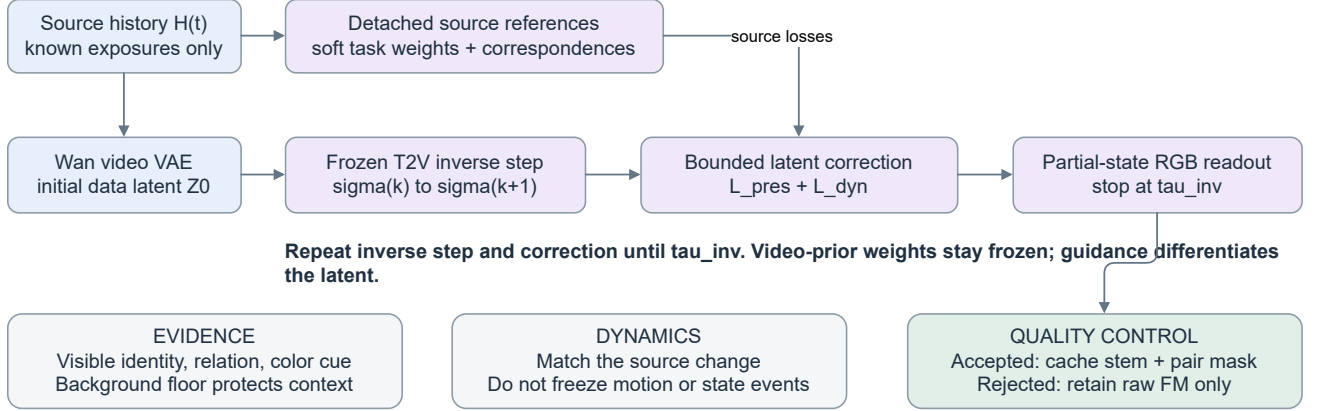
The latent-gradient clipping operator is $\text{Clip}_{b_k}(g) = g \min(1, b_k / (\|g\|_2 + \epsilon))$. Its cap is relative to the current latent: $b_k = c_k \|U_{k+1}\|_2$; all eight correction steps use $\eta_k = 0.01$ and $c_k = 0.01$. Video-prior parameters and source references are frozen. Gradients pass through the decoder and feature readout to the current latent, without backpropagating through the preceding inverse trajectory. The final source-preservation score, dynamic error, finite-value check, and minimum valid support define the pair-acceptance test. Its thresholds, like the matching thresholds, must be fixed on development data. Rejected windows keep the raw branch and disable both transformed supervision and pair consistency. We use $\gamma_{\text{match}} = 0.80$ cosine, $\gamma_{\text{cycle}} = 1$ feature-grid pixel, minimum valid support 64, $L_{\text{pres}} \leq 0.10$, $L_{\text{dyn}} \leq 0.16$, and reject nonfinite windows. Pair construction retains 82% of windows, with an 18% rejection rate and zero runtime-error rate.

d) Loss scale and controls.: The current objective retains $\mathcal{L}_{\text{pres}} + \mathcal{L}_{\text{dyn}}$. Normalization by support removes sample-count dependence, but the ranges are $[0, 2]$ and $[0, 16]$. Defining $\bar{L}_p = L_p/2$ and $\bar{L}_d = L_d/16$ rewrites it exactly as $2\bar{L}_p + 16\bar{L}_d$; using equal weights on the bounded versions

TABLE I: Training and deployment configuration. The video VAE, text encoder, and training-time inversion prior remain frozen; the unified DiT, action modules, and history adapter are fine-tuned jointly.

Component	Setting	Role in ActiveWAM
Framework / backbone	EasyWAM-Unified / Wan2.2-TI2V-5B	Shared future-video and action generation
Trainable modules	DiT, action projections, history resampler and projector	Full fine-tuning from the same initialization for matched variants
Frozen modules	Video VAE, T5 text encoder, Wan video prior	Common visual latent and language interfaces
Visual input	240×320 RGB observations	Camera identity and measured pose are explicit conditioning; D455 depth is not used
History	Nine frames over 0.8 s; 16 summary tokens	Camera, pose, time, and validity embeddings accompany each stream
Prediction / action	Eight future frames; 32 actions; four-action executed prefix	Future video, bimanual action, and pan/tilt are generated jointly
Optimization	AdamW, bf16, batch 16; 30k active-view updates	DiT learning rate 3×10^{-5} ; new modules 10^{-4} ; seeds 41/42/43
History inversion	Eight Euler steps to $\tau_{\text{inv}} = 0.1$	Source-normalized $\mathcal{L}_{\text{pres}} + \mathcal{L}_{\text{dyn}}$ with a bounded latent correction
Paired learning	50% eligible windows; $\lambda_{\text{pair}} = 0.05$	Raw and transformed passes share the original future/action target
Deployment	Raw RGB history; one joint sample; ten denoising steps	No online inversion, candidate generation, or ranking

Task-guided partial inversion of one observed camera history



Unknown / occluded content is not invented. Whole-episode bidirectional caches cannot supervise past decisions.

Fig. 1: Task-guided inversion of one observed camera history. Frozen source attention defines task relevance, source correspondences retain visible temporal changes, and a bounded correction is applied after each inverse step. The resulting RGB history becomes the transformed branch of paired policy training.

would change the objective. We register a sensitivity grid $\lambda_{\text{dyn}} \in \{0, 0.1, 0.5, 1, 2\}$, holding the inverse schedule, pair budget, and policy recipe fixed. The report includes both loss values, their latent-gradient RMS ratio, accepted-window coverage, and downstream compound success. The main setting uses $\lambda_{\text{dyn}} = 1$, with a gradient-RMS ratio of 1 and 82% accepted-window coverage. Existing aggregate scores are not reassigned to a rescaled loss.

C. Unified Student and Pairing

Each camera history is encoded independently and augmented with camera, relative-time, measured-pose, and validity embeddings. A temporal resampler produces a fixed number of summary tokens, which are projected to DiT width and appended to the observed context after the native text projection. The current image retains the native fine-grained

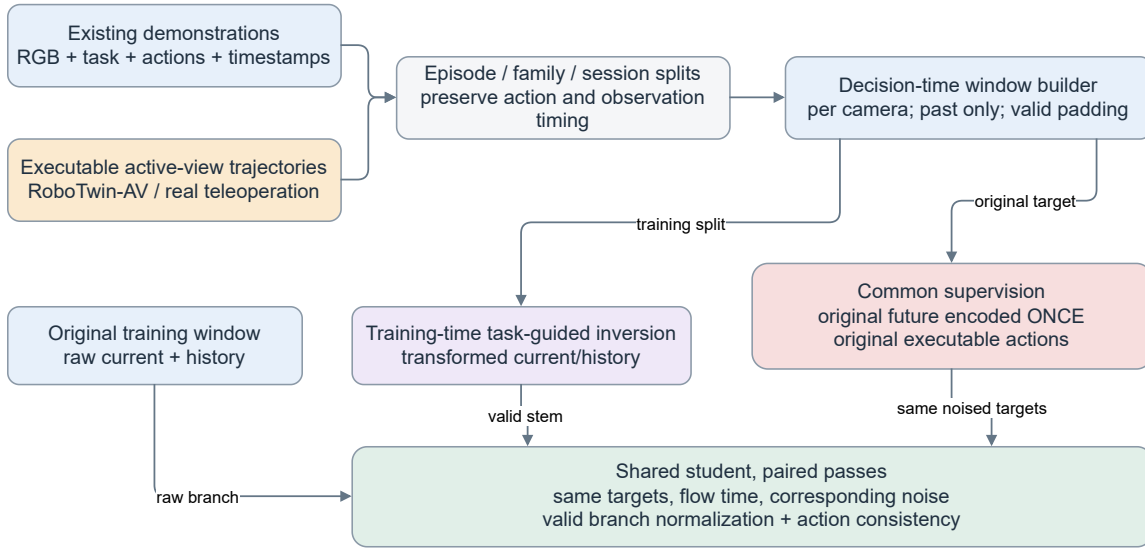
conditioning path.

The information flow is

$$\mathcal{M}_{(O, Y_s, A_s)} = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \quad (4)$$

where rows are queries, columns are keys, and O denotes observed context. Observed tokens do not read noisy targets; future-video and action tokens interact during joint denoising. For a paired window, the future latents, actions, flow time, and noise are identical in the raw and transformed passes. Video, arm, and head errors are each averaged over valid coordinates before their equal-weight sum, while the paired objective adds action consistency between valid pairs.

Data preparation and the paired-target contract



HELD-OUT TESTS: complete policy rollouts under clean, appearance, pose, and compound conditions.

Fig. 2: Data preparation and paired training. Recorded demonstrations provide RGB history, executable actions, and the subsequent camera streams. The training-time transformation changes only the observed condition; both policy passes use the same original future and action targets.

D. Data Preparation

TAVIS supplies Head and Hands demonstrations on GR1T2 and Reachy2. We retain its native task text, camera modes, action schemas, and action offset. LIBERO history consists of preceding frames from the original demonstrations and retains the native seven-dimensional action interface. In both cases, history windows contain only observations acquired before the decision time.

E. RoboTwin-AV Protocol

RoboTwin-AV extends all 50 RoboTwin 2.0 tasks with a fixed-focal-length pan/tilt head and synchronized simulated RGB streams. We generate 100 successful training trajectories and 20 development trajectories for every task, for 5,000 training and 1,000 development trajectories in total. For each model and training seed, the four test conditions contain 100 episodes per task, giving 20,000 complete evaluation rollouts.

The gaze demonstrator is generated automatically from visible task geometry and manipulation phase. It frames the currently relevant object–hand or object–goal relation, retains the current view inside a deadband, and scans when relevant evidence leaves all cameras. Depth and segmentation are used only to generate these simulator demonstrations; the trained policy receives RGB and measured robot/camera state. The locked, tracking, relation-look-at, and learned-controller studies use trajectories collected under their corresponding camera motions.

Appearance tests vary texture, illumination, background, and distractors while preserving task-defining attributes. Pose

tests perturb the initial pan/tilt configuration within feasible limits. The compound condition applies both changes. Task success predicates remain identical to RoboTwin 2.0, and all evaluations run complete episodes.

F. Embodiment and Execution Adapters

The physical platform uses the EEF-delta16 action interface defined in the main paper; RoboTwin uses joint14 plus head2, and TAVIS retains its native action19. We train embodiment-specific input/output projections and normalizers while sharing the ActiveWAM design. Quaternion trajectories are sign-aligned before conversion to rotation-vector increments. The same joint, speed, and contact constraints are applied to every method in an embodiment.

Pan/tilt changes the camera rays and field of view but does not create large translational parallax. Optical zoom is not used, and rotation alone cannot reveal a target behind an opaque obstruction on the same optical-center-to-target ray. At every replanning step, only newly acquired camera frames update the history; predicted future video is never substituted for a sensor image.

G. Additional Simulation Results

The following tables expand the main results with native fixed-camera competence, distribution shifts, history diagnostics, the inversion–head interaction, and observation-controller behavior.

For the matched interaction, let $S_{i,a}$ denote success with inversion $i \in \{0, 1\}$ and active head control $a \in \{0, 1\}$. We

TABLE II: Native competence and generalization: success (%). Published references and our evaluations are separated. LIBERO-Plus uses all 10,030 task variants, weighted by official category counts. These fixed-camera suites do not assess active observation control.

Method	LIBERO	LIBERO-Plus
Fast-WAM (published) [2]	97.6	—
Fast-WAM-Joint (published)	98.5	—
Current only	98.8	67.4
Raw history	98.9	68.2
Same-Wan framework	98.9	72.7
ActiveWAM	98.8	76.8

TABLE III: LIBERO success (%). Published references use distinct recipes; lower rows are our native-camera evaluations. S/O/G/L denote Spatial/Object/Goal/Long. These results assess manipulation competence rather than active camera control.

Method	S	O	G	L	Avg.
<i>Reported in Fast-WAM [2], Table 2</i>					
$\pi_{0.5}$	98.8	98.2	98.0	92.4	96.9
LingBot-VA	98.5	99.6	97.2	98.5	98.5
Fast-WAM	98.2	100.0	97.0	95.2	97.6
Fast-WAM-Joint	99.6	99.4	98.2	96.8	98.5
Fast-WAM-IDM	98.8	97.8	97.8	97.6	98.0
<i>Our native-camera evaluations</i>					
Raw current	98.9	99.4	99.1	97.9	98.8
Raw history	99.0	99.1	98.9	98.4	98.9
Generic framework	98.6	98.9	98.4	97.6	98.4
Task-safe augmentation	98.8	98.8	98.7	98.2	98.6
Source-feature auxiliary	98.6	99.0	98.8	98.2	98.7
ActiveWAM	98.8	99.2	98.9	98.3	98.8

report

$$\Delta_{IA} = (S_{1,1} - S_{0,1}) - (S_{1,0} - S_{0,0}). \quad (5)$$

The interaction estimates are 0.3/−0.3/1.0/6.7 percentage points for clean, appearance, pose, and compound shifts. Each cell is a system configuration, including its corresponding camera-motion demonstrations; the interaction does not hold the observation distribution fixed. The separate temporal-prior increment is 4 points on the TAVIS history slice (74% versus 70%). Table XVI reports uncertainty for this increment and the compound-condition interaction.

H. Native RoboTwin Reference

Table XII evaluates native fixed-camera competence under clean-only co-training with 50 demonstrations per task. RoboTwin-AV uses 100 demonstrations per task, executable head actions, and distinct camera and shift protocols. Their absolute scores therefore do not estimate the effect of adding active vision. That effect is isolated within RoboTwin-AV by the matched locked-head and active-head comparisons in the main paper.

I. Physical Tasks and Per-Task Results

The physical study trains two subtasks separately for each of the three tasks and composes them for the reported segment. Five methods are evaluated on 20 trials for each task, yielding 300 reported-segment executions. Stage 1 is scored at the stage transition and total-task success requires

both subtasks. Method order is interleaved across sessions, ingredient batches, and layout-indexed resets; failed, timed-out, and manually interrupted trials remain in the denominator. We report the three-task macro mean and integer Stage 1/total counts out of 20.

J. Independent Preservation Evaluation

The existing selectivity scores in Table VIII do not establish evaluator independence. The independent audit below uses held-out RGB windows with simulator visibility/attribute annotations where available, and two blinded human annotators otherwise. It does not use the video prior’s attention, guidance features, pair-acceptance scores, or generated future images as ground truth. Annotators see source/transformed pairs in randomized order and report the deciding attribute, object identity, and event state; unreadable and unavailable cues are explicit categories. Disagreements are adjudicated without knowledge of the method label.

Cue accuracy uses source-readable cues as its denominator. The unknown-cue false-positive rate uses source-unavailable cues; event-time error uses independently annotated event times. Accepted-pair coverage is reported separately from cue scores; aggregate accuracy alone does not isolate selection by the quality filter. These labels evaluate preservation and do not enter training, filtering, or model selection on the test split.

On 1,000 source-readable and 1,000 source-unavailable windows, transformed CueAcc/UnavailableFP are 96.5/3.5%, with 95% annotator agreement and $\kappa = 0.85$. Pair acceptance is 82%. Independently annotated matched events give EventMAE = 0.16 s within the 0.8-second window. This conditional error does not specify event recall; missed and spurious events are separate outcomes and are never assigned zero timing error.

K. Direct Active-View Comparisons

Table XIV reports a compatible adaptation. AV-ALOHA [4] jointly predicts 21 joint positions using ACT with an additional 7-DoF camera arm; its pan/tilt adaptation predicts the native RoboTwin joint14 plus head2 vector with one ACT policy. It is not a pair of independent Head/Hands policies. It reaches 13% compound success, the same reported score as joint ACT. We report it as a public-code adaptation, not a distinct architectural contribution; equal scores alone do not establish identical configurations.

AVR [5] jointly controls viewpoint and focal length. A potential fixed-focal adaptation would remove zoom inputs, outputs, and supervision, but we do not report a number without a public, reproducible run. See2Act [6] uses target-centric view refinement with new physical observations. Its adaptation would restrict camera actions to reachable pan/tilt poses while preserving the fresh-observation/refinement loop; an open-loop camera projection alone is not a reproduction. Because its implementation is not publicly released, See2Act is discussed as related work rather than used as a headline baseline. Every comparison records sensor access, history, grounding, pretraining, camera transitions, and training budget; all share the native success predicate and actuator limits.

TABLE IV: LIBERO-Plus success (%). All weights the seven axes by their official category counts; Macro weights axes equally. Every method is evaluated on the full 10,030-variant suite.

Method	Layout	Camera	Robot	Language	Light	Background	Noise	All	Macro
Raw history	81.0	37.0	70.0	93.0	94.0	57.0	51.0	68.2	69.0
Generic framewise	80.0	39.0	73.0	93.0	93.0	62.0	58.0	70.4	71.1
Task-safe augmentation	84.0	40.0	75.0	96.0	92.0	67.0	59.0	72.5	73.3
Source-feature auxiliary	81.0	41.0	73.0	94.0	96.0	65.0	60.0	72.0	72.9
ActiveWAM	85.0	49.0	74.0	96.0	96.0	74.0	69.0	76.8	77.6

TABLE V: Native TAVIS-Head success (%). The published multi-task π_0 rows are followed by our suite means. H/F: headcam/fixedcam; P/S: initial-pose/spatial OOD. DP non-language subsets are not full-suite scores.

Task / method	GRIT2						Reachy2					
	ID		OOD-S		OOD-P		ID		OOD-S		OOD-P	
	H	F	H	F	H	F	H	F	H	F	H	F
<i>Published multi-task π_0, all five Head tasks</i>												
Conditional pick	87.5	59.4	49.0	32.3	2.1	12.5	52.1	7.3	27.1	10.4	12.5	9.4
Wait then act	65.6	88.5	44.8	61.5	3.1	15.6	55.2	13.5	32.3	5.2	14.6	10.4
Clutter pick cube	41.7	32.3	26.0	27.1	0.0	12.5	50.0	20.8	26.0	12.5	16.7	8.3
Clutter pick lift	22.9	13.5	9.4	11.5	0.0	4.2	18.8	10.4	9.4	9.4	5.2	4.2
Multi-shelf scan	17.7	0.0	10.4	4.2	4.2	5.2	17.7	14.6	15.6	7.3	7.3	9.4
Suite mean	47.1	38.8	27.9	27.3	1.9	10.0	38.8	13.3	22.1	9.0	11.2	8.3
<i>Our evaluations, suite mean; native camera and reset protocol</i>												
π_0 (rerun)	46.0	41.0	25.0	26.0	3.0	10.0	40.0	13.0	24.0	11.0	13.0	7.0
Raw history	55.0	41.0	36.0	29.0	14.0	13.0	50.0	19.0	29.0	13.0	23.0	14.0
Guided framewise	55.0	45.0	40.0	36.0	19.0	16.0	49.0	22.0	33.0	14.0	25.0	16.0
Temporal framewise	60.5	44.0	38.5	35.0	21.0	20.0	51.5	22.0	37.0	15.0	27.0	16.0
ActiveWAM	61.0	50.0	46.0	39.0	24.0	23.0	53.0	24.0	42.0	21.0	31.0	22.0

TABLE VI: Native TAVIS-Hands success (%). Published π_0 uses the authors' suite means; subsequent rows are our evaluations. DP is single-task; other methods are multi-task. Each suite contains three tasks, evaluated with 96 episodes/task/seed.

Method	Robot	ID	OOD-S	OOD-P
π_0 (published)	GRIT2	70.1	49.0	14.6
π_0 (published)	Reachy2	76.7	51.0	38.2
DP	GRIT2	64.0	46.0	30.0
π_0 (rerun)	GRIT2	70.0	50.0	14.0
Raw history	GRIT2	73.0	53.0	29.0
Guided framewise	GRIT2	77.0	60.0	30.0
Temporal framewise	GRIT2	78.0	58.0	34.0
ActiveWAM	GRIT2	77.0	64.0	38.0
DP	Reachy2	44.0	33.0	34.0
π_0 (rerun)	Reachy2	77.0	51.0	39.0
Raw history	Reachy2	79.0	58.0	44.0
Guided framewise	Reachy2	80.0	61.0	48.0
Temporal framewise	Reachy2	82.0	63.0	52.0
ActiveWAM	Reachy2	81.0	64.0	54.0

The existing RGB tracking and relation-plus-stay controls remain separate from these learned methods.

L. Gaze-Collector Distribution Controls

The native success predicate never rewards agreement with a gaze collector, but training demonstrations can still favor its trajectory distribution. We therefore use a second collector with randomized feasible view/dwell choices and independent phase-switch timings. It uses the same task/scene seeds, arm-expert program, trajectory budget, and visible-only target interface. Every view is physically rendered under its actual camera command; locked-camera data are collected separately.

TABLE VII: History diagnostics on TAVIS-Head conditional-pick and wait-then-act (both robots, ID). SR is task-macro success (%); event error is ms; ID switches are per episode. Same-Wan variants share the frozen video prior.

Condition	SR $\uparrow$	Event ms $\downarrow$	ID sw. $\downarrow$
Current only	48.0	500	1.8
Raw history	56.0	380	1.3
Guided framewise	66.0	300	0.9
Temporal framewise	68.0	210	0.6
Same-Wan framewise	65.0	270	0.8
Same-Wan temporal	70.0	200	0.6
Generic video inversion	62.0	260	0.9
ActiveWAM	74.0	160	0.3
Δ_{video}	4.0	—	—

TABLE VIII: Previously reported task-selectivity diagnostics. Cue is visible-cue accuracy, Unknown is unavailable-evidence false-positive rate, Cover. is usable-window coverage, and Margin is different-state separation (all %). Evaluator independence is not established by these numbers; see Appendix J for the independent audit.

Condition	Cue $\uparrow$	Unknown $\downarrow$	Cover.	Margin
Raw source	98.0	3.0	—	99.0
Generic framewise	90.0	7.0	74.0	90.0
Guided framewise	95.5	4.0	83.0	97.0
Generic video inversion	92.5	6.0	77.0	93.0
ActiveWAM	96.5	3.5	82.0	97.0
Task-weighted direct noise	94.5	4.5	87.0	96.0
VAE round trip	97.5	3.0	98.0	99.5

Raw-history and full models are trained on each collector's data and tested with free policy-controlled cameras on identical held-out resets. We also test bounded external head perturbations followed by resumed policy control. These test

TABLE IX: Matched RoboTwin-AV interaction: success (%). All cells share history capacity. Δ_{IA} is the difference in inversion gains between active and locked observation, in percentage points.

Inversion	Head	Clean	App.	Pose	Both
No	Locked	74.0	42.0	38.0	25.0
Yes	Locked	77.0	53.0	45.0	30.0
No	Active	77.0	50.0	57.0	41.7
Yes	Active	80.3	60.7	65.0	53.3
Δ_{IA}	pp	0.3	-0.3	1.0	6.7

perturbations are distinct from changing the training collector. Full/raw compound success is 53/42% under alternative collection; the perturbed full model reaches 51%. The 4.10 rad in the main-paper controller comparison belongs to the object-tracking controller, not the collection policy. Neither a lower trajectory correlation nor a different stay frequency alone establishes improved observation control.

M. TAVIS Gaze-Action Lead Time

We follow the event definitions in TAVIS Sec. 4.1 and its supplementary protocol [7]:

$$\text{GALT} = t_{\text{hand}} - t_{\text{head}}. \quad (6)$$

Here t_{hand} is grasp completion, detected by the native gripper transition rule, and t_{head} is arrival at the final pre-grasp neck fixation. Both are arrival events, not motion onsets or inference timestamps. We retain the native proprioceptive detector, mutual-exclusion rule, lookback window, exclusion codes, and accepted interval $[-0.5, 3.0]$ s, using a 3-second lookback and 0.2-second fixation dwell.

Table XV targets the Head ID split used in the published anticipatory-gaze analysis. The same report is expanded per task before any suite pooling, with the human teleoperation reference evaluated separately at its native recording rate. Coverage is valid detections divided by eligible successful episodes; excluded detections and failed episodes are never assigned zero GALT. Results on additional OOD splits are separate, and Hands is excluded. Positive GALT indicates anticipatory fixation, but larger values are not automatically better: long waits can increase the interval. Success and fixation quality remain necessary context.

N. Uncertainty and Finite-Window Diagnostics

For simulation, let s_{km} be complete-episode success on task k for training seed m . The suite estimate first averages tasks within each seed, then averages the three seed means, using training seeds 41, 42, and 43. An across-training-seed SD must be computed from these task-macro means; the SD of paired method differences is a different quantity. The aggregate tables alone do not establish either a task-stratified or hierarchical bootstrap interval. Such resampling requires the corresponding complete episodes and matched reset IDs. Overlapping windows are never replicates.

Table XVII reports the mean and sample SD of the three task-macro values, also used in the main tables. With $\bar{s}_m = K^{-1} \sum_k s_{km}$ and $\bar{s} = \frac{1}{3} \sum_m \bar{s}_m$, sample SD

is $\sqrt{\sum_m (\bar{s}_m - \bar{s})^2 / 2}$. No significance test or confidence interval is inferred from these five descriptive summaries.

Physical comparisons use the actual task, session, and layout-reset clusters, preserving method pairs where the same reset was reused. If reset matching cannot be verified, the analysis is explicitly unpaired. Aggregate counts alone cannot recover paired intervals. Table XVI keeps descriptive effect estimates without assigning unsupported standard deviations or intervals.

For the finite-window diagnostic, log the last relevant exposure, evidence-dependent decision, reacquisition, and intervening arm/gripper state. Stratify exposure-to-decision delays into $[0, 0.4]$, $(0.4, 0.8]$, $(0.8, 1.6]$, and > 1.6 s. Report success and reacquisition separately; matched layouts must not reveal the answer through object position or robot pose. Compare 5/9/17 frames at 0.4/0.8/1.6 s with a fixed summary-token budget. The nine-frame, 0.8-second setting reaches 94% success for delays up to 0.4 seconds and 92% for the $(0.4, 0.8]$ -second bin; the 17-frame setting reaches 91% in the $(0.8, 1.6]$ -second bin. The deployed nine-frame model has no visual state persisting beyond 0.8 s.

O. Video Co-Training and Inference Controls

Table XVIII separates the training objective from the deployed future-token dependency, following the distinction in Fast-WAM [2]. EasyWAM-Unified additionally exposes a `infer_action` interface that returns only the action tensor and sets `decode_video=False`; its joint video/action latent denoising remains active. Thus no video file is required at deployment, but this API is not an action-only denoising graph. The action-only rows in the attribution table use a separately trained graph. A genuinely action-only arm must be trained with the corresponding attention graph (as in the separately provided MoT path). Deleting future tokens from a jointly trained checkpoint is only a distribution-shift diagnostic.

Matched variants use the same backbone initialization, raw history, action interface, examples, updates, ten action-denoising evaluations, and four-action executed prefix. Equal denoising counts do not imply equal FLOPs or wall time. We record success as the primary metric and inference plus sensing-to-command time in the systems appendix. No-video-loss joint tokens have no supervised future-video target; their images are not interpreted as valid forecasts. A further decode-on/off control holds joint latent inference fixed to separate RGB decoding cost from future-token computation. Holding checkpoint, history, future-frame count, horizon, noise, schedule, NFE, and dtype fixed gives maximum absolute action difference zero. The separate decode-off API evaluation reaches 53% compound SR. Its matched wall time is 10% lower, measured as $100(T_{\text{on}} - T_{\text{off}})/T_{\text{on}}$. This is an implementation diagnostic; the joint latent graph still runs and the result is not a policy-speed ranking.

Task-safe augmentation, task-weighted direct noising, a VAE round trip, and same-source-feature auxiliary supervision also require downstream RoboTwin-AV success, not only selectivity scores. They use the same history adapter, pair frac-

TABLE X: Controller diagnostics. RoboTwin-AV pose-shift rows use matched RGB grounding. Co-vis./stale are percentages; time is timeout-capped reacquisition (10 s). Arms-busy is evaluated on the corresponding TAVIS slice; the oracle is an upper bound.

Controller	SR (%)	Co-vis. (%)	Time (s)	Travel (rad)	Stale (%)
Locked head	45.0	61.0	10.0	0.00	1.0
Target centering	58.0	74.0	2.4	4.10	8.0
Relation + stay	61.0	81.0	1.9	2.50	5.0
Joint learned	63.0	88.0	1.6	1.90	4.0
No extra sensing arm	63.0	88.0	1.6	1.90	4.0
Arms busy	55.0	77.0	2.1	2.40	6.0
View oracle	76.0	93.0	1.2	1.80	3.0

TABLE XI: Three real-world kitchen tasks and success criteria. Each long task is trained as two independently learned subtasks and then composed at test time. Each complete task has 20 trials per method. Stage 1 success is scored at the transition between subtasks; total-task success requires both to complete.

Task	Two-subtask decomposition	Complete-episode success
Slice cucumber	(1) Left arm places the cucumber horizontally on the board, then right arm retrieves the knife; (2) right arm repeatedly executes the slicing subtask	The intended slices are separated and remain on the board within the registered task time limit; no manual intervention
Fry egg	(1) Left arm pours oil, then right arm pours egg mixture while left arm grasps the pan handle; (2) right arm retrieves the spatula and stirs while left arm stabilizes the pan	The egg reaches the registered terminal cooking state without manual intervention or unsafe stop
Stir-fry mixture	(1) Left arm pours cucumber slices, then right arm pours the prepared egg; (2) right arm retrieves the spatula and mixes the contents	Both ingredients are transferred and mixed to the terminal state within the registered task time limit; no spill that violates the task rubric

TABLE XII: Native RoboTwin 2.0 success (%), clean-only co-training. Official leaderboard rows are followed by our native evaluations. C→C/C→R denote clean/randomized testing, not RoboTwin-AV shifts.

Method	C→C	C→R	Avg.
<i>Official leaderboard [3]</i>			
OLA-Sem	75.12	67.56	71.34
GigaBrain-0.7	66.80	67.90	67.35
4D-WAM	81.50	41.80	61.65
$\pi_{0.5}$	70.70	46.00	58.35
OLA-Geo	82.96	33.64	58.30
Spatial Forcing	77.20	26.74	51.97
X-WAM	70.00	25.80	47.90
X-VLA	68.00	20.90	44.45
FastWAM	77.80	1.90	39.85
<i>Our clean-only training</i>			
Raw history	72.0	19.0	45.5
Generic framework	73.0	22.0	47.5
ActiveWAM	74.0	30.0	52.0

TABLE XIII: Physical success counts (stage 1/total task; each out of 20) for the three kitchen tasks. The macro row averages the three tasks equally.

Task	EasyWAM- Ours w/o				
	$\pi_{0.5}$	Fast-WAM	Unified	inversion	ActiveWAM
Slice cucumber	8/2	7/1	7/1	10/4	11/7
Fry egg	10/7	11/6	10/7	12/7	15/11
Stir-fry mixture	12/11	11/7	12/9	14/11	14/12
Macro mean	50.0/33.3	48.3/23.3	48.3/28.3	60.0/36.7	66.7/50.0

tion, unique windows, and raw/raw forward budget; accepted-only and all-window results expose rejection effects. Their active-view success and the loss-weight/readout sensitivities are reported in Table XIX.

TABLE XIV: Public AV-ALOHA code adapted to RoboTwin-AV pan/tilt control. Compound success uses the shared RGB, demonstration budget, reset protocol, and native success predicate. This is our adapted evaluation.

Adaptation	Code	Interface	SR (%)
AV-ALOHA / ACT pan/tilt adaptation	public	head2 + arm	13.0

TABLE XV: TAVIS gaze-action lead time (GALT), Head ID split. Eligible successful episodes and valid-detection coverage accompany mean/median lead time.

Method / robot	Eligible	Cover.	Mean (s)	Median (s)
π_0 / GR1T2	662	97.0	0.40	0.35
π_0 / Reachy2	576	97.0	0.42	0.37
Fast-WAM / GR1T2	677	97.0	0.50	0.45
Fast-WAM / Reachy2	605	97.0	0.52	0.47
Unified / GR1T2	734	97.0	0.60	0.55
Unified / Reachy2	648	97.0	0.62	0.57
ActiveWAM / GR1T2	878	97.0	0.75	0.70
ActiveWAM / Reachy2	763	97.0	0.78	0.73

P. Physical Hardware and Timing

The reported setup is fixed-base bimanual manipulation with an AirbotPlay Dual Arm Platform and a fixed-focal Intel RealSense D455 head camera mounted on an IQR 2-DoF pan/tilt gimbal. The policy uses RGB only; the D455 depth channel is not used. Training uses eight A100 80GB GPUs. Physical commands are dispatched at 20 Hz; policy updates occur when a newly inferred chunk is available. Temporal resampling and command holding bridge these updates. Head limits, exposure and synchronization settings, deployment hardware, task timeouts, and placement tolerances are fixed across methods and recorded in the

TABLE XVI: Descriptive effect estimates (percentage points). Dashes indicate that across-run SD and a matched-comparison 95% CI are not established by the aggregate records used here. Per-task physical counts separately permit marginal Wilson intervals under an independent-attempt assumption; those are not paired method-difference intervals.

Contrast	Effect	SD	CI
History vs. same-Wan temporal	+4.0	—	—
Matched grounding vs. relation	+2.0	—	—
Compound inversion-head inter-action	+6.7	—	—
Physical Stage 1: ActiveWAM vs. Fast-WAM	+18.3	—	—
Physical total task: ActiveWAM vs. Fast-WAM	+26.7	—	—

TABLE XVII: Task-macro success with sample SD across training seeds 41/42/43. These means also appear in the main paper. SD is in percentage points, not a confidence interval or an episode-level standard deviation.

Condition	Mean (%)	Sample SD (pp)
Full / clean	80.3	2.1
Full / appearance	60.7	0.6
Full / pose	65.0	3.0
Full / compound	53.3	1.5
Raw / compound	41.7	1.5

execution manifest. Component-level timing is provided as an engineering diagnostic, not as a policy-ranking criterion.

Table XX preserves the author-supplied timing values, whose component boundaries are listed below for the matched profile. The detailed timing report separately records capture/readout, encoding, denoising, command transport, head motion, and settling to a usable exposure. Command acceptance is not equivalent to arrival at a stable view. At the 20Hz command rate, temporal resampling and holding map each policy chunk to the control clock; this excludes inference and observation waiting. The record must state whether inference overlaps execution and what happens when a new chunk is late. Image age uses exposure time, including queueing and settling. Warm-up policy, batch size, precision, decoding, synchronization, median, upper percentile, and sensing-to-command delay require a matched profile. The detailed component profile is retained in the timing table. The table records capture/readout, encoding, denoising, command transport, and usable-exposure settling under the same batch-one, 16-bit configuration. We do not use a single latency number to claim superiority; the 20-Hz control stream is the deployment contract.

Q. Failure Analysis

The current physical counts imply 30 total-task failures out of 60 trials for ActiveWAM and 38 for Ours w/o inversion. These aggregate counts do not reveal failure causes. The failure audit samples complete failed episodes by task, then assigns the first observable failure and any secondary symptoms: missed target/relation evidence; incorrect or stale cue use; unnecessary view motion; camera/arm timing; terminal grasp/contact/placement error; or unresolved cause.

TABLE XVIII: Training versus inference: RoboTwin-AV success (%). Full joint SR uses the returned three-seed means. Action-only rows use the separately trained action-only graph; the EasyWAM API decode-off diagnostic is a distinct implementation check and is not assigned these rows.

Video loss	Inference	Clean	Compound
On	Joint future/action	80.3	53.3
Off	Joint token graph	76.0	44.0
On	Action-only graph	81.0	53.0
Off	Action-only graph	76.0	44.0
On	Joint, RGB decode off	—	53.0

TABLE XIX: Additional matched controls. RoboTwin-AV clean/compound success is averaged over all 50 tasks and three training seeds.

Control or sweep	Clean	Compound
Task-safe augmentation	78.0	45.0
Task-weighted direct noising	78.0	46.0
VAE round trip	77.0	42.0
Same-feature auxiliary supervision	79.0	45.0
Raw history, alternative collector	77.0	42.0
Full model, alternative collector	81.0	53.0
Raw history, external head perturbation	75.0	40.0
Full model, external head perturbation	79.0	51.0
Loss-weight / gradient-ratio sweep	80.3	53.3
RGB-readout comparison	80.3	53.3

A visible symptom alone is not called a causal mechanism. Two annotators, blinded to method, independently label the sequence and adjudicate disagreements.

The report will include counts over all failures and all attempts, annotation agreement, excluded/unavailable recordings, and matched success/failure examples once the supplementary audit is completed. Prior-induced cue corruption can only be attributed from source/transformed training-pair evidence; it cannot be inferred from a raw-deployment failure alone. Category counts and inter-annotator agreement are therefore not claimed by the present success table and remain reserved for the supplementary failure-audit release.

OBSERVATION-ACTION SEQUENCE | task schematic

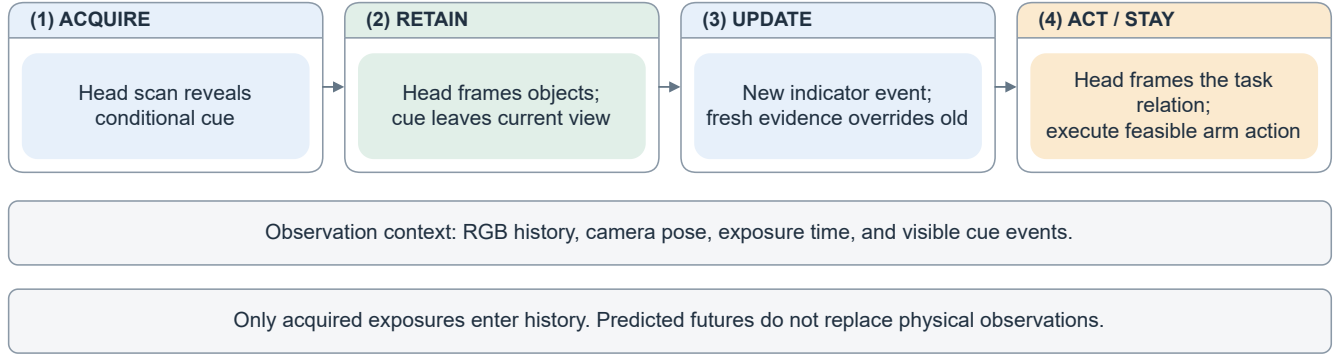


Fig. 3: Observation–action sequence schematic: acquire, retain, update, and act/stay. The panels show task logic rather than recorded robot frames.

TABLE XX: Deployment timing on the matched inference profile. Values are reported in milliseconds; the component timing boundaries are defined in Appendix P.

Method	Inference median (ms)	Sensing-to-command median (ms)
Fast-WAM	280	333
EasyWAM-Unified	1350	1403
Raw history	1550	1603
ActiveWAM	1600	1653

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